# The second law

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Explain the direction of heat transfer and the limits of energy conversion.

**SUCCESS CRITERIA - I can...**

1. Predict spontaneous net heat flow.
2. Explain why moving heat from cold to hot needs an additional process.
3. Distinguish energy conservation from process direction.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “spontaneous” and “heat” to explain your model or result. Sentence starters are optional.

**spontaneous**: able to occur without the specified external work input

**heat**: energy transferred because of a temperature difference

**work**: energy transferred by a force along a displacement

## 01 Warm-up: bring back an earlier idea

A 0.3 kg metal cools from 91 to 31°C while 0.25 kg water warms from 11 to 31°C.
**Use** water c=4200; neglect cup/losses.
**Find** metal c.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $Q_{hot} = Q_{cold} + W_{in}$.
Refrigerator cycle: heat amounts are positive magnitudes. Work input moves heat from cold to hot; spontaneous flow is hot to cold.

**Worked reasoning:** Hot and cold blocks in contact exchange energy until thermal equilibrium. Total energy can be conserved while the temperature difference and ability to drive a heat engine decrease.

**Your turn: Draw** before-and-after temperature states and the spontaneous heat-flow direction.

**Sentence starter:** The ... in my model represents ... . Its label shows ... .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

**Marzano check - circle:** ① ② ③

# The second law

**antocciphysics**



Continue down each column.

Sentence starters are optional. You may plan in any language.

## 03 Read and interpret the model

A 50°C block touches a 10°C block.
**Draw** net heat flow and predict their trend toward equilibrium.

**Sentence starter:** I predict … . My model supports this because … .



50°C



10°C

Each outline represents the labeled sample or container.

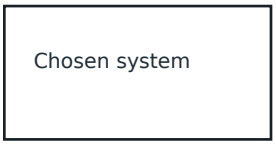
**Marzano check - circle:** 1 2 3

## 04 Apply the physics

During one modeled refrigerator cycle, 200 J is removed from the cold compartment using 50 J of electrical work.
**Draw** the energy flows.
**Find** the energy released to the room and the coefficient of performance, $Q_c/W$.
**Explain** why COP is not heat-engine efficiency.

**Sentence starter:** Under the assumption …, the evidence … supports … because … .

Draw and label all energy transfers across the boundary.




Stored energy change: _____ J

**Marzano check - circle:** 1 2 3

# The second law

**antocciphysics**



Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Explain your reasoning

A proposed machine takes heat from one reservoir and turns all of it into work in a repeating cycle. What thermodynamic limitation is missing?

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

A 0.4 kg object moving at 6 m/s stops through friction over 0.8 m.
**Find** energy transformed and average opposing-force magnitude.
**Show** the energy changes and transfers; label units.

**Word help:** magnitude = size of a quantity without its direction

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Initial** **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

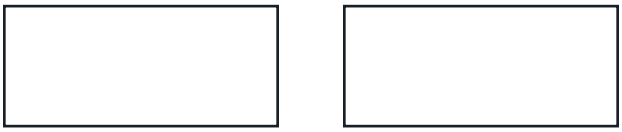
**Marzano check - circle:** 1 2 3

## 07 Exit check

Two bodies exchange heat in an isolated system and reach the same temperature. Is the total energy lost? What happens to the ability to extract work from their temperature difference?

**Sentence starter:** Both ... . They differ in ... because ... .

Sample A Sample B



Label temperatures and draw energy-transfer arrows.

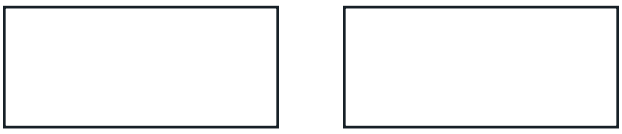
**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

A refrigerator removes 200 J from a cold space while receiving 80 J electrical work.
**Find** heat delivered to the room and explain why it is larger than 200 J.

**Sentence starter:** Both ... . They differ in ... because ... .

Sample A Sample B



Label temperatures and draw energy-transfer arrows.

**Marzano check - circle:** 1 2 3